

Empiric Antibiotics by Site of Infection

IM chalk-talk quick reference · community vs. hospital acquisition drives the "bugs" · always narrow to cultures · adapt to local antibiogram.
Adapted from TeachIM (CC BY-NC 4.0).

Pneumonia		
Type	Bugs	Empiric drugs
CAP	S. pneumo, GNR (H.flu, Morax), atypicals	Ceftriaxone + azithro/doxy, or resp FQ
HAP/VAP (>48h)	+ MRSA, + Pseudomonas, + MDR GNR	(Vanc/linezolid) + (Zosyn/cefepime/mero/aztreonam)
Aspiration	+ oral anaerobes	CAP cov; anaerobes <i>only</i> if abscess/empyema

- **HCAP retired (2019).** Add MRSA/PsA only if prior MRSA/PsA or hosp + IV abx <90 d.
- MRSA nares PCR neg → stop vancomycin.
- **2025 ATS:** shorter courses (3–5 d if stable by 48–72 h); hydrocortisone <400 mg/d ×5–7 d for severe CAP only (CAPE COD).

Urinary Tract	
Syndrome	Drugs
Cystitis (E. coli 75–95%)	Nitrofurantoin ×5 d / TMP-SMX ×3 d (R<20%) / fosfomycin ×1
Pyelo / complicated	Ceftriaxone → Zosyn/mero if MDR risk or critically ill; + vanc/dapto if Enterococcus

- **NEVER** nitrofurantoin/fosfomycin for pyelonephritis (bladder-only).
- **CAUTI** = sx/signs + bacteriuria + catheter (or ≤48 h). Adds PsA, Enterococcus. **Don't treat asymptomatic bacteriuria.**
- MDR risk (<3 mo): prior MDR/ESBL, care facility, recent FQ/Bactrim/broad BL, high-MDR travel.

Skin & Soft Tissue		
Type	Bug	Drugs
Nonpurulent	Strep >> MSSA	Cephalexin / cefazolin / ceftriaxone
Purulent	MRSA > MSSA	I&D first + vanc/dapto, or PO Bactrim/doxy
NSTI	GAS, S.aureus, GNR, Clostridium	STAT surgery + vanc/dapto + Zosyn/mero + clinda

- Purulent = drainage problem first; culture pus, not superficial swabs.
- **NSTI red flags:** pain/tenderness beyond erythema, rapid spread, toxicity, crepitus, bullae, duskiness → OR, don't delay.
- Clindamycin = antitoxin (shuts off exotoxin) in NSTI / strep TSS.

Intra-abdominal		
Setting	Bugs	Drugs
Community	Enteric GNR, strep, anaerobes (B. frag)	3rd-gen ceph + metronidazole (or FQ + metro)
Hospital/ HCA	+ Pseudomonas, + Enterococcus, ± MRSA	Zosyn / (cefepime+metro) / meropenem ± vanc

- **Source control is half the treatment** (drain / cholecystectomy / repair).
- Avoid amp-sulbactam empirically (high E. coli resistance).

The "Hospital Trio"
Across every site, healthcare/hospital exposure adds MRSA , Pseudomonas , and Enterococcus . Escalate empirically only with risk factors; de-escalate by culture at 48–72 h .
Hospital-acquired triggers: onset >48 h after admission; hospitalization ≥48 h or care-facility stay within 90 d; broad-spectrum abx >5 d in prior 90 d.

Do / Avoid
• DO draw cultures before/with first dose; narrow promptly. • DO use a respiratory FQ as a single-agent CAP option when beta-lactam+macrolide isn't feasible. • AVOID reflexive anaerobic coverage in aspiration (only abscess/empyema). • AVOID treating asymptomatic bacteriuria (except pregnancy / pre-urologic procedure). • AVOID antibiotics-without-surgery in NSTI.